Casting Processes

Recall the concepts of Casting in EMATL 1:

Expendable mold casting; sand casting, shell mould casting

Permanent mold casting: die, centrifugal, and investment

to pick the casting method, consider metal type, scale, shape and size, dimensional accuracy, and surface finish.

Pattern & mould

- prepares mold cavity, bottom pattern called drag, top called cope.
- mould consists of pouring basin, sprue, runner, gate, into the casting, + riser.
- cavities are made with cores
- Mold material picked needs to have refractory characteristics, and can widthstand the pouring temp, either with sand (expendable mold), or metal/graphite (permanent mold)

metal grains are finer closer to the mold walls, then columnar grains after due to directional cooling (anisotropic strength), then equiaxed grains.

some shrinkage compensation needed, either for uniform shrinkage (causing smaller grains), or non uniform shrinkage from differential cooling, causing stresses, warping and cracking (differential cooling due to geometry or external conditions)

shrinkage should not be obstructed by mould material other casting defects will occur.

casting is processed with shot blasting to remove sand, breaking excess material by manual methods or grinding/sawing, then cleaned with shot blasting or chemical methods.

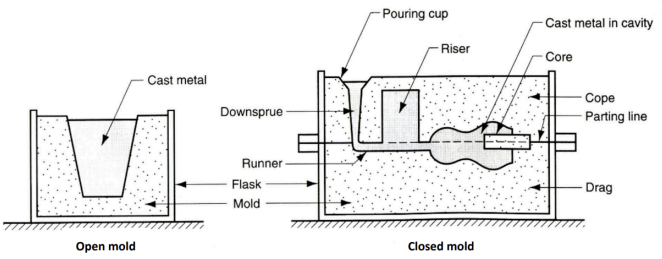
heat treatment may be done.

QC check is done after

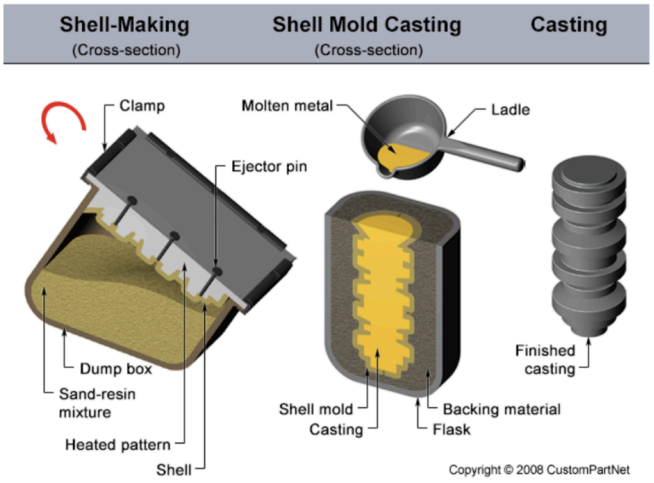
Sand Casting

most common, using pouring basin >> sprue >> runner >> gate >> casting, with riser. (for closed molds)

has cope and drag with core if necessary (for closed molds)



Shell Casting (new)



Generally, process is by curing sand onto pattern to form shell as mould, then molten metal can be poured and casting is made.

Full process:

1. fine silica sand mixed with thermosetting phenolic resin and liquid catalyst is applied (dumped, blown or shot) to hot pattern (cast iron at 230 to 315C)
2. sand resin mixture sits on pattern to partially cure
3. excess sand is removed to form 10 to 20mm thickness shell mould
4. pattern and shell is placed in oven to cure fully, then remove shell (tensile strength of 2.4 to 3.1MPa)
5. glue shells with thermoset adhesive

6. place shell mould in flask to allow metal to be poured.

advantages

9mm shell thickness (?? shrinkage)

smoother shell mould than sand >> good surface of ~2.5um

easy metal flow

good dimmensional accuracy

economical even at large scale production

disadvantages

expensive metal pattern needed

less viable for small scale

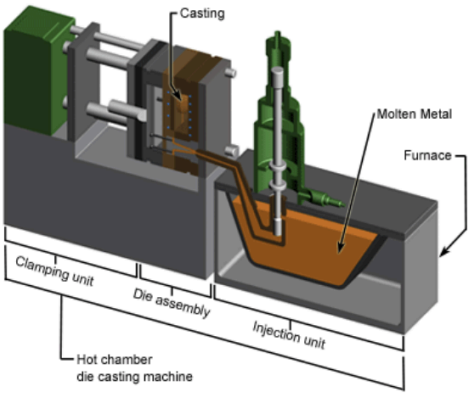
feasible on small/medium size parts only

suitable for casting of >10kg

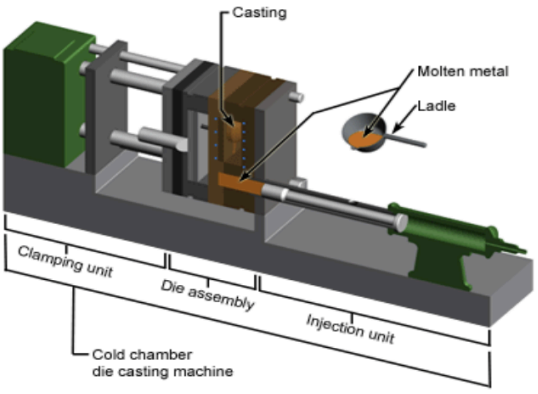
used for gears, valve bodies, bushings and cam shafts

Die Casting

Recall the two types of die casting, hot chamber and cold chamber:



Hot Chamber Machine



Cold Chamber Machine

advantages:

high speed at large scale

duplicates intricate details

no pattern needed

- long mould life of 100K cycles
- disadvantages:
  - expensive machinery and mould
  - extreme heat limited by mould >> select few metals viable like Zn, Cu, Al, etc.
  - high pressure needed >> larger parts not viable

Investment Casting

- recall the process, involving
- creating the pattern
  - wax pattern made with central gate running into castings, tree like assembly.
- creating the mould
  - pattern dipped into ceramic slurry to form ceramic shell, repeated until thick enough, then heated in oven to be poured out to form hollow shell
- pouring
  - mold heated to furnace at 1000C, then filled with metal to fill cavity, sometimes vaccum or pressure needed.
- Cooling and finishing
  - casting is cooled and mold is broken out for part to be seperated from gating and post processed.
- advantages
  - high complexity viable
  - good dimensional accuracy (+-0.075mm)
  - good surface finish
  - reusable wax pattern after melting
  - minimal further machining
- disadvantages
  - casting < 5kg
  - max dimension usually <300mm
  - thickness of part restricted to >15mm

Centrifugal Casting (new)

permanent cylidrinical mould made to be rotated at 300 to 3000rpm, then metal poured in cavity to conform to rotating mold

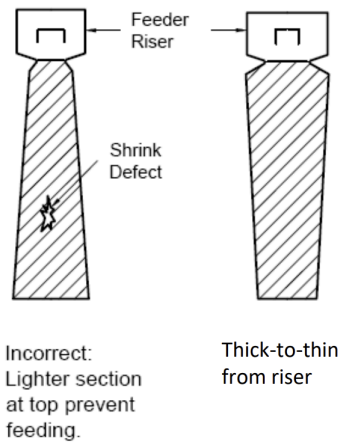
higher density and outer most regions, with good surface finish

good for pipes, bushings, gears and flywheels.

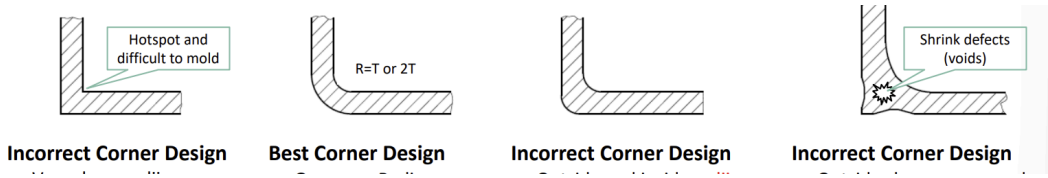
Design for Casting

Some methods to make designs better for casting include:

- add risers so that dimensions increase in dimension progressively to riser to offset shrinkage



- maximise filets to reduce stresses, mitigate cracks, tears and draws, and make corners moldable (by eliminating hot spots)
  - use uniform wall thickness, with R = T or 2T, otherwise stress concentrations, voids or sinks occur.



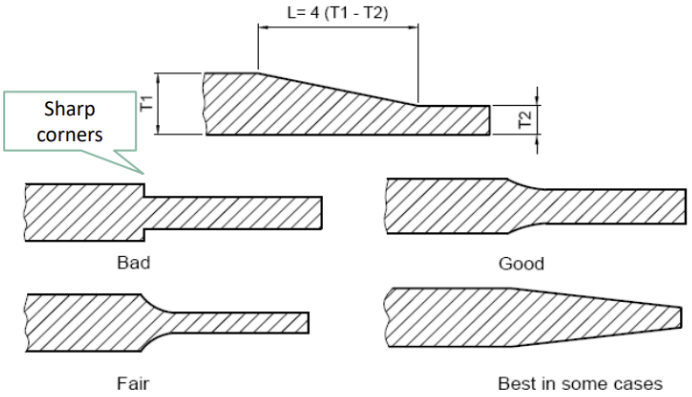
- add drafts outward on surfaces perpendicular to parting line for easier ejection of part.
  - 1 deg for external surfaces like walls, 2 deg for internal surfaces

- bring adjoining sections together to minimise overaccumulation of metal (melt)

- stagger ribs to allow uniform cooling to reduce distortion

Staggered ribs reduce distortion caused by shrinkage.

- use uniform wall thickness for uniform cooling >> reduce defects
  - if needed, thicker sections should be adjacent to risers so that shrinkage is compensated.
  - thinner wall thicknesses shorten cycle time and part volume (better)
- avoid abrupt section changes (stress concentrations)
  - thicknesses should not excess 2:1 ratio
  - if <2:1, use filets, otherwise, use tapers



- design ribs well
  - avoid shallow ribs, too far spaced out ribs, or thin ribs.
  - apply base rib filets
  - use radial ribs if possible
- disgusing sink marks (due to ribs)
  - use grooves, beads, or a surface texture at the opposite end of ribs
- design bosses and pads
  - use bosses of height less than thickness, with tapers or filets at the boss base
  - exception for thin casting sections (T is
- avoid undercuts
  - change mold seam to remove mold undercuts
  - redesign part to remove undercuts
- straighten parting lines
  - depends on shape of casting and mould, but ideally use straight parting lines
- drill small holes
- account for shrinkage

Casting Defects

surface defects

penetration: high fluidity of metal penetrates mould

scab: upheaved sand falls from mould surface and metal flows around sand and mold

blow: large surface cavity due to gases displacing metal on convex surface

scar: shallow blow on flat surface

blister: scar but with a thin layer of metal covering it, due to excessive gas forming from sand

buckle: v shaped depression caused by expansion of sand mould

- internal defects
  - sand blow & pinholes: gas cavity below surface/many small gas cavities
  - cold shot: metal splatters durign pouring and solid metal globules form and are entrapped in casting.
- visible defects
  - misrun: metal solidifed before filling cavity
  - cold shut: metal flowed but failed to fuse
  - shrinkage cavity: depression in surface
  - shift: misalignment of mold or core